Student Performance Analytics System

Project Documentation – Python with AI (EWB Courses)

1. Project Title

Project Title	Student Performance Analytics System
Student Name	Dnyaneshwari Maruti Pacharane
Course	Python with AI, EWB Courses
College / Batch	Institute of Business Management & Rural Development Ahmednagar 2025-2027
Submission Date	15 October 2026

2. Objective

When I started this project, I thought about how a teacher checks marks. It is usually done in a notebook or an Excel sheet, and working out totals, grades and results for every student takes a lot of time. It is also easy to make small mistakes.

So I made a Python program that does this work automatically. It reads the student data from a file, calculates the totals, averages and grades, shows who passed and who failed, and finds the best subject and the top students. It also points out students who may need extra help because of low marks or low attendance. While making it, I practised functions, loops, NumPy and Pandas, which we learned in class.

3. Technologies Used

- **Python 3:** the language I used to write the whole project.
- **NumPy:** for calculations like sum, mean, median, standard deviation, maximum and minimum.
- **Pandas:** to read the CSV file and to sort, filter and group the data.
- **VS Code:** where I wrote and ran my code.
- **Git and GitHub:** to save the project and submit it online.

4. Dataset Description

I made a file called `students.csv` with **20 students** from five departments: Computer Science, Information Technology, Data Science, Mechanical and Electronics. Each student has marks in five subjects and an attendance percentage. The columns are explained below.

Column	Description

Column	Description
Student_ID	Unique ID of each student (S001 to S020)
Name	Student name
Department	Department of the student
Python	Marks in Python (out of 100)
Maths	Marks in Maths (out of 100)
Statistics	Marks in Statistics (out of 100)
Data_Structures	Marks in Data Structures (out of 100)
English	Marks in English (out of 100)
Attendance	Attendance percentage

There are no missing values in the data. I created the data myself, but I tried to keep it realistic by including strong students, average students and a few weak ones, so that the results would be interesting.

5. Implementation

My project has three main files. `main.py` runs the program and prints the report, `functions.py` keeps all the reusable functions, and `students.csv` holds the data.

Step 1: Loading the data

First I load the data with a function called `load_data()`. It uses `pd.read_csv()` to put the CSV file into a Pandas DataFrame. Then the program prints the number of records, the column names, the missing values and the first five rows, so I can check that the data loaded properly.

Step 2: Finding the subject columns

Next, `get_subject_columns()` picks out the subject columns by leaving out ID, name, department and attendance. This way, if I add a new subject to the CSV later, I do not have to change the code.

Step 3: Processing each student

The main calculations happen in `calculate_student_metrics()`. It converts the marks into a NumPy array and then:

- finds each student's **total** with `np.sum(marks, axis=1)`
- finds the **average** with `np.mean(marks, axis=1)`
- gives a **grade** by calling `assign_grade()` for every student
- decides **Pass or Fail** using `np.where()` and `np.all()`
- marks the **attendance** as OK or Low

Step 4: Grading and pass/fail rules

Average marks	Grade
90 and above	A+
80 to 89	A
70 to 79	B
60 to 69	C
50 to 59	D
40 to 49	E
Below 40	F

- **Pass:** a student must get at least 40 marks in every subject. If even one subject is below 40, the result is Fail.
- **Low attendance:** attendance below 75% is marked as Low. I use it only as a warning, so it does not change pass or fail.

Step 5: Analysis

- **Class summary:** the overall average, median, standard deviation, highest and lowest average, and how many students passed and failed.
- **Grade distribution:** a loop goes through the grades A+ to F and prints a small bar using `#` symbols.
- **Subject-wise analysis:** `subject_analysis()` loops through each subject and finds its average, highest marks, lowest marks and pass percentage.
- **Top performers:** `top_performers()` sorts the students by average marks and gives the top five.
- **Department analysis:** `department_analysis()` uses `groupby()` to compare the departments.
- **Students needing attention:** students who failed or have low attendance are filtered into a separate list.

Step 6: Saving the output

At the end, the complete table with totals, grades and results is saved to `results.csv` using `to_csv()`.

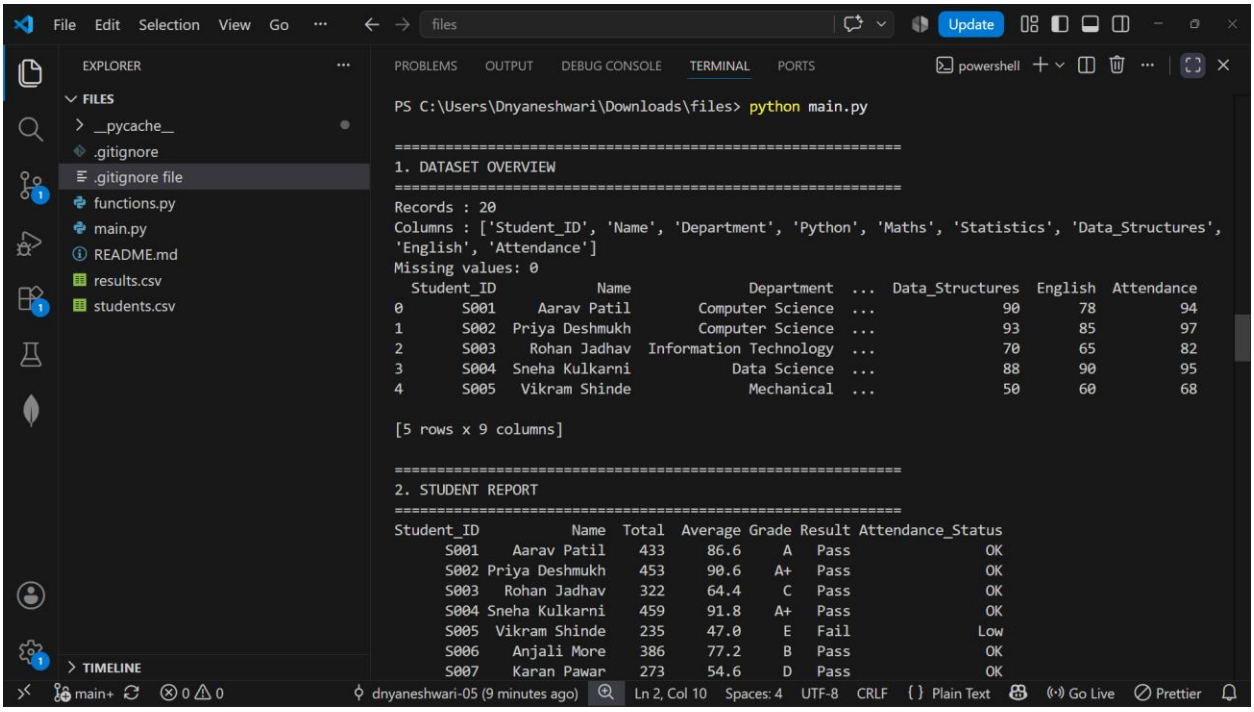
6. Key Features

- Reads the student data from a CSV file using Pandas
- Calculates total and average marks for each student
- Gives grades based on a clear grading rule
- Shows who passed and who failed
- Finds the class average, median, highest and lowest
- Shows the grade distribution
- Shows subject-wise performance with the best and the weakest subject
- Lists the top five students

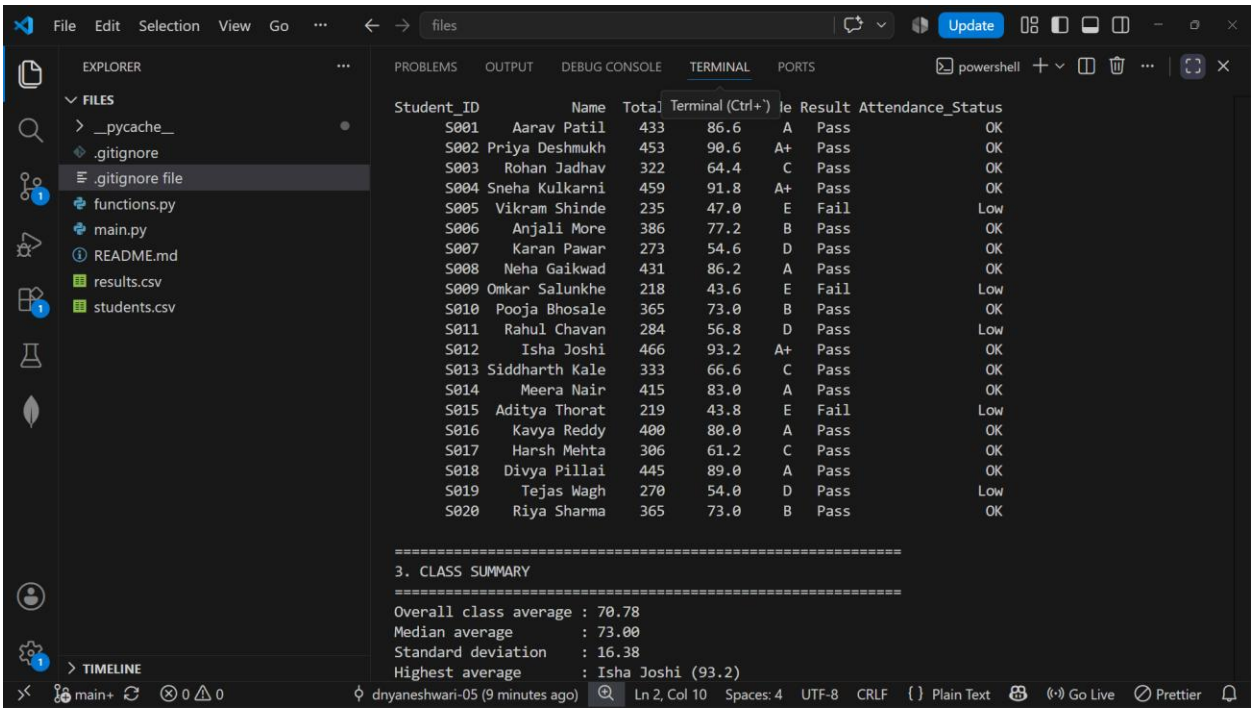
- Compares the departments
- Lists students with a failing result or low attendance
- Uses reusable functions, loops and NumPy operations
- Saves the final results in `results.csv`

7. Output / Screenshots

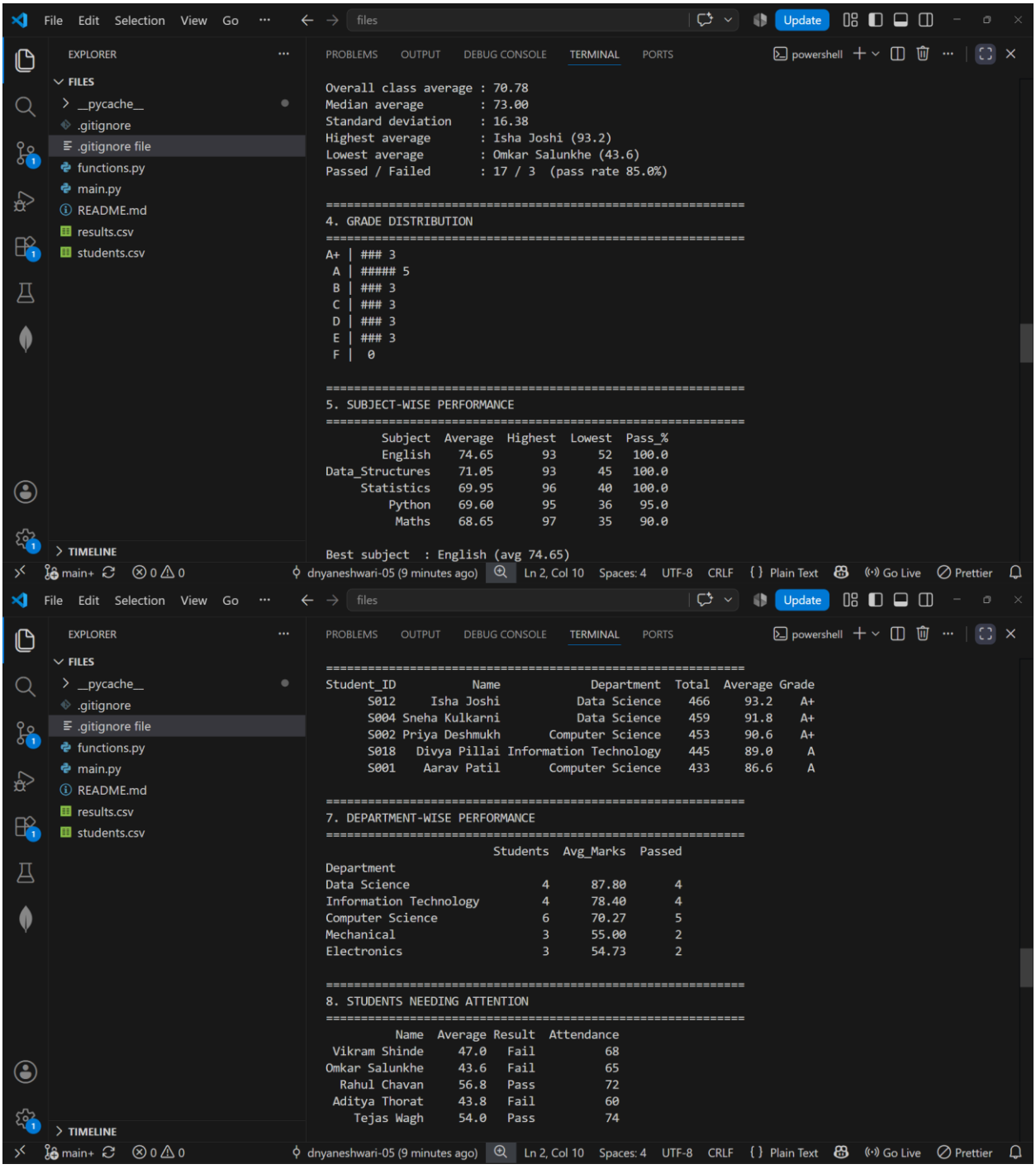
1.



2.


File Edit Selection View Go
files
Update
EXPLORER
FILES
__pycache__
.gitignore
.gitignore file
functions.py
main.py
README.md
results.csv
students.csv
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
Terminal (Ctrl+`)
Student_ID Name Total le Result Attendance_Status
S001 Aarav Patil 433 86.6 A Pass OK
S002 Priya Deshmukh 453 90.6 A+ Pass OK
S003 Rohan Jadhav 322 64.4 C Pass OK
S004 Sneha Kulkarni 459 91.8 A+ Pass OK
S005 Vikram Shinde 235 47.0 E Fail Low
S006 Anjali More 386 77.2 B Pass OK
S007 Karan Pawar 273 54.6 D Pass OK
S008 Neha Gaikwad 431 86.2 A Pass OK
S009 Omkar Salunkhe 218 43.6 E Fail Low
S010 Pooja Bhosale 365 73.0 B Pass OK
S011 Rahul Chavan 284 56.8 D Pass Low
S012 Isha Joshi 466 93.2 A+ Pass OK
S013 Siddharth Kale 333 66.6 C Pass OK
S014 Meera Nair 415 83.0 A Pass OK
S015 Aditya Thorat 219 43.8 E Fail Low
S016 Kavya Reddy 400 80.0 A Pass OK
S017 Harsh Mehta 306 61.2 C Pass OK
S018 Divya Pillai 445 89.0 A Pass OK
S019 Tejas Wagh 270 54.0 D Pass Low
S020 Riya Sharma 365 73.0 B Pass OK
3. CLASS SUMMARY
Overall class average : 70.78
Median average : 73.00
Standard deviation : 16.38
Highest average : Isha Joshi (93.2)
main+
dnyaneshwari-05 (9 minutes ago)
Ln 2, Col 10 Spaces: 4 UTF-8 CRLF Plain Text Go Live Prettier

3.


File Edit Selection View Go
files
Update
EXPLORER
FILES
__pycache__
.gitignore
.gitignore file
functions.py
main.py
README.md
results.csv
students.csv
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
Overall class average : 70.78
Median average : 73.00
Standard deviation : 16.38
Highest average : Isha Joshi (93.2)
Lowest average : Omkar Salunkhe (43.6)
Passed / Failed : 17 / 3 (pass rate 85.0%)
4. GRADE DISTRIBUTION
A+ | ### 3
A | ##### 5
B | ### 3
C | ### 3
D | ### 3
E | ### 3
F | 0
5. SUBJECT-WISE PERFORMANCE
Subject Average Highest Lowest Pass_%
English 74.65 93 52 100.0
Data_Structures 71.05 93 45 100.0
Statistics 69.95 96 40 100.0
Python 69.60 95 36 95.0
Maths 68.65 97 35 90.0
Best subject : English (avg 74.65)
main+ 0 0 dnyaneshwari-05 (9 minutes ago) Ln 2, Col 10 Spaces: 4 UTF-8 CRLF Plain Text Go Live Prettier
File Edit Selection View Go
files
Update
EXPLORER
FILES
__pycache__
.gitignore
.gitignore file
functions.py
main.py
README.md
results.csv
students.csv
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
Student_ID Name Department Total Average Grade
S012 Isha Joshi Data Science 466 93.2 A+
S004 Sneha Kulkarni Data Science 459 91.8 A+
S002 Priya Deshmukh Computer Science 453 90.6 A+
S018 Divya Pillai Information Technology 445 89.0 A
S001 Aarav Patil Computer Science 433 86.6 A
7. DEPARTMENT-WISE PERFORMANCE
Students Avg_Marks Passed
Department
Data Science 4 87.80 4
Information Technology 4 78.40 4
Computer Science 6 70.27 5
Mechanical 3 55.00 2
Electronics 3 54.73 2
8. STUDENTS NEEDING ATTENTION
Name Average Result Attendance
Vikram Shinde 47.0 Fail 68
Omkar Salunkhe 43.6 Fail 65
Rahul Chavan 56.8 Pass 72
Aditya Thorat 43.8 Fail 60
Tejas Wagh 54.0 Pass 74
main+ 0 0 dnyaneshwari-05 (9 minutes ago) Ln 2, Col 10 Spaces: 4 UTF-8 CRLF Plain Text Go Live Prettier

Key results from the program:

Result	Value
Total students	20
Overall class average	70.78
Highest average	Isha Joshi (93.2)
Lowest average	Omkar Salunkhe (43.6)
Passed / Failed	17 / 3 (pass rate 85%)
Best subject	English (average 74.65)
Weakest subject	Maths (average 68.65)
Best department	Data Science (average 87.80)

Top 5 performers

Rank	Name	Average	Grade
1	Isha Joshi	93.2	A+
2	Sneha Kulkarni	91.8	A+
3	Priya Deshmukh	90.6	A+
4	Divya Pillai	89.0	A
5	Aarav Patil	86.6	A

8. Final Outcome

My project works as a complete system. It reads the data of 20 students, does all the calculations and prints a clear report that anyone can understand without reading the code.

From the results, the overall class average is 70.78 and 17 out of 20 students passed, so the pass rate is 85%. English is the strongest subject and Maths is the weakest. Data Science students have the best department average, while Mechanical and Electronics are the lowest. The three students who failed (Vikram Shinde, Omkar Salunkhe and Aditya Thorat) also have low attendance, so attendance may be connected to the results. The program lists these students so a teacher can help them early.

Overall, the project covers all the required features of the assignment and shows how functions, loops, NumPy and Pandas can be used together to solve a real problem.

9. Challenges and Learning

Challenges I faced and how I solved them

- **No Run button in VS Code:** VS Code was in Restricted Mode, which blocks running code. I trusted the folder and installed the Python extension from Microsoft, and then it worked.
- **Error while adding the GitHub link:** I pasted only the repository URL in the terminal and got an error. Then I understood that the URL has to be used inside the command `git remote add origin <URL>`.
- **Push was rejected:** GitHub had already made its own README, so the two histories did not match. I checked that only that README would be replaced and then used a force push.
- **Wrong file name and cache files:** I had saved the ignore file as `.gitignore file`, so Git did not ignore the `__pycache__` folder. I renamed it to `.gitignore` and removed the cache files from Git.

I learned

- How to split a project into a main file, a functions file and a data file
- How to write functions and use them from another file
- How NumPy does calculations on a whole array at once, for example `np.mean(marks, axis=1)` and `np.where()`
- How to load, sort, filter and group data with Pandas
- How to publish a project using Git and GitHub
- How to read an error message and fix the problem step by step

10. GitHub Repository

https://github.com/dnyaneshwari-05/student-performance-analytics

The repository has `main.py`, `functions.py`, `students.csv`, `results.csv`, `README.md` and this documentation.